not be numbers. For example in the random experiment of tossing a coin, the outcomes are $H$ (head) or $T$ (tail). It is necessary to deal with numerical values, in some situation, for outcomes of random experiment. Therefore, we assign a number to each outcome of the experiment say 1to head and 0 to tail. Such an assignment of numerical values to the elements in $S$ is called a *random variable*. *A random variable is a function*. Thus, a random variable is:

**Definition 11.1**

**A random variable** $X$ *is a function defined on a sample space* $S$ *into the real numbers* $\mathbb{R}$ *such that the inverse image of points or subset or interval of* $\mathbb{R}$ *is an event in* $S$, *for which probability is assigned.*

We use the capital letters of the alphabet, such as *X, Y,* and $Z$ to represent the random variables and the small letters, such as *x, y,* and $z$ to represent the possible values of the random variables.

Suppose $S = \{\omega_1, \omega_2, \omega_3, \ldots\}$ is the sample space of a random experiment and $\mathbb{R}$ denotes the real line. Then the random variable $X$ is a real valued function defined on $S$ and is denoted by $X : S \to \mathbb{R}$. If $\omega$ is a sample point in $S$, then $X(\omega)$ is a real number.

The range set is the collection of $X(\omega)$ such that $\omega \in S$. That is the range set denoted by $R_x$ is $R_x = \{X(\omega) / \omega \in S\}$.

Fig 11.1 shows the mapping of some sample points $\omega_i$ or events of the *Sample space* $S$ on the *real line R*.

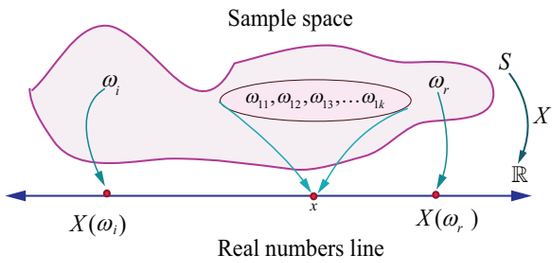


Fig. 11.1

For instance, if $x$ is a possible value of $X$ for $\omega_{11}, \omega_{12}, \omega_{13}, \ldots \omega_{1k} \in S$, then $\{\omega_{11}, \omega_{12}, \omega_{13}, \ldots \omega_{1k}\}$ is called inverse image of $x$.

That is $X^{-1}(x) = \{\omega_{11}, \omega_{12}, \omega_{13}, \ldots \omega_{1k}\}$ is an event in $S$

**Illustration 11.1**

Suppose a coin is tossed once. The sample space consists of two sample points $H$ (head) and $T$ (tail).

That is $S = \{T, H\}$

Let $X : S \to \mathbb{R}$ be the number of heads

Then $X(T) = 0$, and $X(H) = 1$.

Thus $X$ is a random variable that takes on the values 0 and 1. If $X(\omega)$ denotes the number of heads, then

$$X(\omega) = \begin{cases} 0 & \text{for } \omega = \text{Tail} \\ 1 & \text{for } \omega = \text{Head} \end{cases}$$

**Example 11.1**

Suppose two coins are tossed once. If $X$ denotes the number of tails, (i) write down the sample space (ii) find the inverse image of 1 (iii) the values of the random variable and number of elements in its inverse images.

**Solution**

(i) The sample space $S = \{H, T\} \times \{H, T\}$



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